Saumya Khubchandani

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EDUCATION

B.Tech. in Computer Science and Engineering (Information Security)
Vellore Institute of Technology (VIT), Vellore | CGPA: 8.1/10.0 | 2023 – 2027
Class 12th (ISC): 94% (2023) | Class 10th (ICSE): 89% (2021)

TECHNICAL SKILLS

Languages: Java, Python, TypeScript, SQL, JavaScript, C++, C
Backend and Databases: FastAPI, Node.js, Express.js, Next.js, PostgreSQL, Drizzle ORM, Supabase, DynamoDB, REST APIs
Payments and Security: Razorpay integration, HMAC-SHA256 webhook verification, idempotent payment processing, OTP authentication, rate limiting, role-based access control
DevOps and Testing: Docker, GitHub Actions, CI/CD, Git, JUnit, Playwright, Vercel, AWS API Gateway
Core Concepts: Data Structures and Algorithms, Object-Oriented Programming, Database Design, API Design, Idempotency and Retries, Information Security
AI/ML: PyTorch, OpenCV, YOLOv8, Hugging Face Transformers, NumPy, Pandas

EXPERIENCE

AI/ML Engineer Intern **Jun 2026 – Aug 2026**
Data Science Technologies Pvt. Ltd., Remote

- Built the FastAPI video-processing backend and React dashboard for a football tactical analysis platform (player tracking, heatmaps, match insights).
- Shipped 3 processing tiers (3-minute GPU preview to 35-minute full CPU run), letting clients trade speed for accuracy.
- Reduced distance-measurement error from 68% to 3.2% on the SoccerNet-GSR benchmark using a YOLOv8, ByteTrack, SigLIP and homography pipeline.
- Cut tracker-ID fragmentation by 77% (109 IDs to 25 real players) using SigLIP re-identification and offline tracklet refinement.

Tech stack: Python, FastAPI, PyTorch, OpenCV, YOLOv8, React.js

Quant Development Intern **May 2025 – Jul 2025**
AceQuant Research, Mumbai

- Integrated 2 broker APIs (AngelOne, Zerodha) to generate performance metrics, dashboards and datasets for strategy evaluation.
- Built and deployed 4+ algorithmic trading strategies (SMA, EMA, Bollinger Bands, MACD), validated on live market data across 10+ segments.

Tech stack: Python, NumPy, Matplotlib, AngelOne API, Zerodha API

PROJECTS

saas-sync: Incremental ELT Engine | github.com/saumyak0405/saas-sync **Oct 2026**

- Built a Java engine syncing Stripe and GitHub data incrementally into PostgreSQL behind a single pluggable connector interface.
- Made syncs crash-safe and idempotent with transactional checkpoints, primary-key upserts, rate-limit-aware retries and automatic schema evolution; 61 tests incl. PostgreSQL integration.

Tech stack: Java, PostgreSQL, JUnit, Docker, GitHub Actions

Seva: Verified Domestic-Worker Marketplace | seva-iota.vercel.app **Sep 2026 – Present**

- Integrated Razorpay payments with HMAC-SHA256 webhook verification, idempotent processing and automatic refunds.
- Secured login with hashed one-time passwords, per-phone and per-IP rate limiting and role-based access control.
- Built and deployed a bilingual (Hindi/English) marketplace with 3 user roles and 8 service categories for households in Agra.
- Wrote 65 Playwright end-to-end tests and set up CI/CD from GitHub to Vercel with a Supabase PostgreSQL database.

Tech stack: Next.js, TypeScript, PostgreSQL, Drizzle ORM, Razorpay, Playwright, Vercel

LEADERSHIP

Vice Chairperson **Mar 2025 – Present**
IEEE Professional Communication Society, VIT Vellore

- Increased member engagement by 40% through recruitment drives and organized technical events for 500+ students.

CERTIFICATIONS

- Mastering Data Structures and Algorithms using C++ (Udemy)
- Quantitative Finance: Algorithmic Trading in Python (Udemy)